

Final Project Requirement

Overview

MXET 300 is focused on mechatronics system using the intelligent robot as an illustration. The MXET 300 Lab intends to cover the basics of mobile robot by taking you through the process of constructing the SCUTTLE robot.

The MXET 300 Final Project is an opportunity to put your new robotics skills into practice.

Rules

1. Teams for the final project should have be made out of 4 students within their lab sections. 3 students team is fine if cannot be grouped in 4.
2. Each team should have a YouTube channel.
3. Final Project should have some level of autonomy and machine intelligence (execute mission without user intervention)

References

The Following information can be found in the [official SCUTTLE website](#). Most of the SCUTTLE related videos are located in the following YouTube Playlist: [SCUTTLE videos](#). Here are some of the most critical:

1. [Flash a Debian Image](#)
2. [Linux Root, Home, directories](#)
3. [Connect by USB & find SSID](#)
4. [Intro to SCUTTLE software architecture](#)
5. [How to NodeRed with BeagleBone?](#)
6. [Anatomy of DuPont Connector](#)
7. [Build a Power Splitter](#)
8. [How does I2C works](#)
9. [Send & Receive MQTT messages, display with NodeRed on BeagleBone Blue](#)

The following are a list of the most visited webpages used through the entire lab. This list is useful as it contains all the information needed about SCUTTLE, the MXET 300 lab and class.

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1. [Scutte ALL DOCUMENTS](#) ← Please Visit!
2. [Wiring Diagram](#)
3. [SCUTTLE Kinematics](#)
4. [SCUTTLE home page](#)
5. [SCUTTLE software architecture](#)
6. [Scutte IoTGuide](#)
7. [As5048A Datasheet](#)
8. [SCUTTLE GITHUB](#)

Requirements

1. Each project must have a basic module. The basic module can be either of two options: obstacle avoidance or SLAM (simultaneous localization and map building). The sensors can be either LIDAR or stereo camera.

** First choose a basic module you would like to go with. If you choose the obstacle avoidance module, needs to avoid dynamic obstacle reliably.

** Do a comprehensive literature survey on what open source codes are available for the basic function you select. The literature survey needs to be comprehensive and well documented in the project report.

** Choose the best open source code, and then implement it to the scuttle robot. Explain why you choose that open source and its pros and cons in the report. Being able to implement a state of the art open source code and achieve good performance is preferred option and a plus in grading. (+++)

** If the open source code you select is too computationally intensive and cannot be handled by the Beaglebone, we have Rasberry Pi or Jetson Nano in the lab you can use.

** If you cannot make the open source code work, then it is still acceptable to write your own code for your basic module. You need to make sure that the code is well written and achieve good performance.

2. Extended Application based on the basic module

** The first basic module must be well completed before moving on to the second part.

** The second part is an application that the team can propose based on the basic module. There can be extra hardware added if needed.

**** Examples can be a serving robot, cleaning robot, golf ball collector, etc. You can be creative on this part.**

Upcoming Deadlines

1. Final Project Proposal (DUE by Nov 3rd midnight)

This should be a YouTube video with a maximum duration of 5 minutes, where each final project team will explain the selected idea for the final project and describe the plan.

- Submit video link on canvas with team member names.
- Each team only needs to submit one proposal.
- Submit Power Point Slides used for video on canvas.
- Dr. Song will perform a Q&A to all teams during lecture time.
- Slides should have the mission selected, planned effort distribution among team members and anticipated timeline.

2. Mid term literature survey report (DUE: TBD)

A mid term report on the survey of open source modules of the basic function you select need to be submitted about 2 weeks after the proposal. The report needs to be comprehensive and list all the potential options. Classify those options two categories: one is ROS dependent and the other is not dependent on ROS. List pros and cons of each open source option and explain why you proceed with the particular one you choose.

- Each team only needs to submit one report.

3. Final Report and Demo (DUE: Last lecture)

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- Each team needs to submit one report, one YouTube presentation link with demo video, and one copy of slides for the presentation.
- The demo videos should be long enough to show the performance of the robot in multiple environments. The demo should be designed in a way to easily show the reliability, accuracy and performance of robot.
- The performance of the basic module should also be highlighted in the demo.
- There should be a section of effort distribution of individual team members in the presentation as well as the final report.